PRAKTIKUM 4
SINGLY LINKED LIST (Percobaan 5-8)

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D4 TEKNIK KOMPUTER
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
2024

5. **Percobaan 5 – Menghapus Seluruh Node Pada Singly Linked List**

Pada percobaan ini, seluruh node dalam SLL dapat dihapus secara iterative dengan menggunakan fungsi deleteList() pada 2 jenis pemrograman, menggunakan C dan C++.

a. **Menggunakan C**

```
// C program to delete a linked list
#include<stdio.h>
#include<stdlib.h>
#include<assert.h>

/* Link list node */
struct Node
{
	int data;
	struct Node* next;
};

/* Function to delete the entire linked list */
void deleteList(struct Node** head_ref)
{
/* deref head_ref to get the real head */
struct Node* current = *head_ref;
struct Node* next;

while (current != NULL)
{
	next = current->next;
	free(current);
	current = next;
}

/* deref head_ref to affect the real head back
	in the caller. */
*head_ref = NULL;
}

/* Given a reference (pointer to pointer) to the head
of a list and an int, push a new node on the front
of the list. */
void push(struct Node** head_ref, int new_data)
{
```

```
	/* allocate node */
	struct Node* new_node =
			(struct Node*) malloc(sizeof(struct Node));

	/* put in the data */
	new_node->data = new_data;

	/* link the old list off the new node */
	new_node->next = (*head_ref);

	/* move the head to point to the new node */
	(*head_ref) = new_node;
}

/* Driver program to test count function*/
int main()
{
	/* Start with the empty list */
	struct Node* head = NULL;

	/* Use push() to construct below list
	1->12->1->4->1 */
	push(&head, 1);
	push(&head, 4);
	push(&head, 1);
	push(&head, 12);
	push(&head, 1);

	printf("\n Deleting linked list");
	deleteList(&head);

	printf("\n Linked list deleted");
}
```

Hasil percobaan:

```c
#include <stdio.h>
#include <stdlib.h>
struct Node
{
    int data;
    struct Node* next;
};
void deleteList(struct Node** head_ref)
{
    struct Node* current = *head_ref;
    struct Node* next;
    while (current != NULL)
    {
        next = current->next;
        free(current);
        current = next;
    }
    *head_ref = NULL;
}
void push(struct Node** head_ref, int new_data)
{
    struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}
int main()
{
    struct Node* head = NULL;
    push(&head, 1);
    push(&head, 4);
    push(&head, 1);
    push(&head, 12);
    push(&head, 1);

    printf("\n Deleting linked list");
    deleteList(&head);
    printf("\n Linked list deleted");
    return 0;
}
```

```
Windows PowerShell
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PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-nni12fde.qjc' '--stdout=Microsoft-MIEngine-Out-sqnexreq.rbf' '--stderr=Microsoft-MIEngine-Error-tqj2ttmp.uin' '--pid=Microsoft-MIEngine-Pid-ozlz3lu4.der' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'

 Deleting linked list
 Linked list deleted
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-tgyo1ou2.dvp' '--stdout=Microsoft-MIEngine-Out-ebj5cfuz.zo4' '--stderr=Microsoft-MIEngine-Error-kmze4esi.xzm' '--pid=Microsoft-MIEngine-Pid-ldzxgxl2.gtl' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'

 Deleting linked list
 Linked list deleted
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

```
 Deleting linked list
 Linked list deleted
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

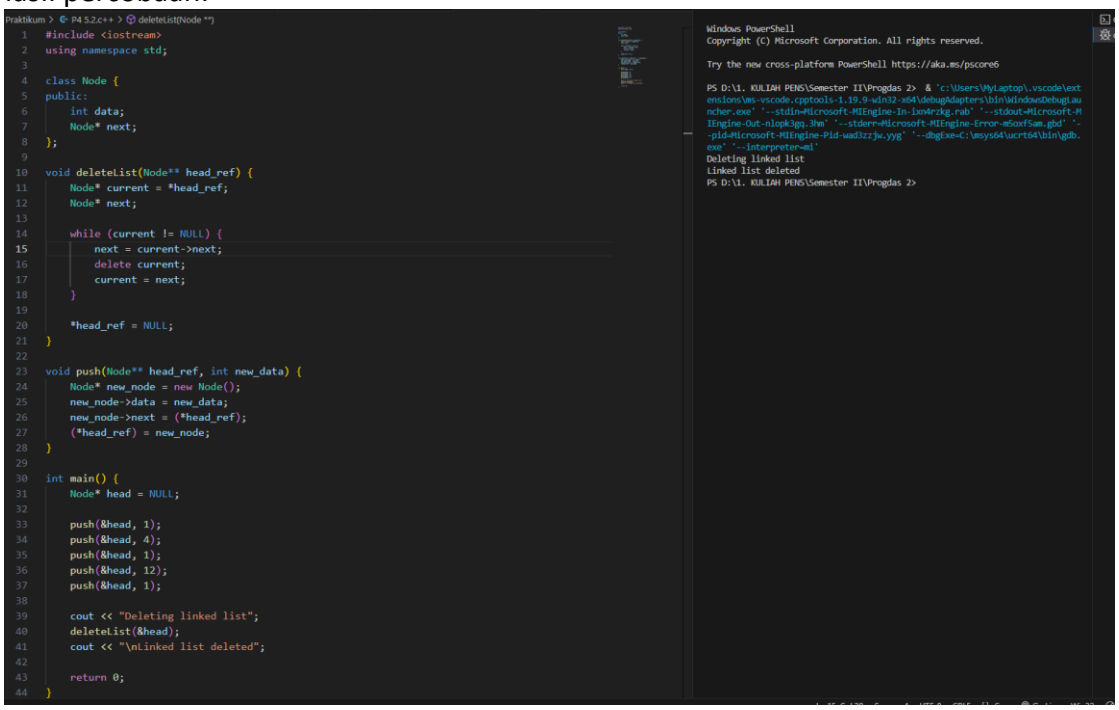
b. Menggunakan C++

```
// C++ program to delete a linked list
#include <bits/stdc++.h>
using namespace std;

/* Link list node */
class Node
```

```
{
        public:
        int data;
        Node* next;
};

/* Function to delete the entire linked list */
void deleteList(Node** head_ref)
{

/* deref head_ref to get the real head */
Node* current = *head_ref;
Node* next;

while (current != NULL)
{
        next = current->next;
        free(current);
        current = next;
}

/* deref head_ref to affect the real head back
        in the caller. */
*head_ref = NULL;
}

/* Given a reference (pointer to pointer) to the head
of a list and an int, push a new node on the front
of the list. */
void push(Node** head_ref, int new_data)
{
        /* allocate node */
        Node* new_node = new Node();

        /* put in the data */
        new_node->data = new_data;

        /* link the old list off the new node */
        new_node->next = (*head_ref);

        /* move the head to point to the new node */
```

```
        (*head_ref) = new_node;
}

/* Driver code*/
int main()
{
        /* Start with the empty list */
        Node* head = NULL;

        /* Use push() to construct below list
        1->12->1->4->1 */
        push(&head, 1);
        push(&head, 4);
        push(&head, 1);
        push(&head, 12);
        push(&head, 1);

        cout << "Deleting linked list";
        deleteList(&head);

        cout << "\nLinked list deleted";
}
```

Hasil percobaan:




```
Deleting linked list
Linked list deleted
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

6. **Percobaan 6 – Menemukan Panjang Singly Linked List**

Pada percobaan ini, panjang atau jumlah keseluruhan node dari SLL akan dihitung menggunakan 2 teknik, yaitu iterative dan recursive.

a. **Iterative**

```
// Iterative C++ program to find length
// or count of nodes in a linked list
#include <bits/stdc++.h>
using namespace std;

/* Link list node */
class Node
{
        public:
        int data;
        Node* next;
};
          /* Given a reference (pointer to pointer)
          to the headof a list and an int, push a new
          node on the front
          of the list. */
          void push(Node** head_ref, int new_data)
          {
                  /* allocate node */
                  Node* new_node =new Node();

                  /* put in the data
                  */ new_node->data
                  = new_data;

                  /* link the old list off the
                  new node */new_node-
                  >next = (*head_ref);

                  /* move the head to point to the
                  new node */(*head_ref) =
                  new_node;
          }
```

```
/* Counts no. of nodes in linked list */
int getCount(Node* head)
{
        int count = 0; // Initialize count
        Node* current = head; // Initialize current
        while (current != NULL)
        {
                count++;
                current = current->next;
        }
        return count;
}

/* Driver program to test count function*/
int main()
{
        /* Start with the empty list */
        Node* head = NULL;

        /* Use push() to construct below list
        1->2->1->3->1 */
        push(&head, 1);
        push(&head, 3); push(&head, 1);
        push(&head, 2);
        push(&head, 1);

      /* Check the count function */
      cout<<"count of nodes is "<< getCount(head);
return 0;
}
```

Hasil percobaan

```
Praktikum > P4 6.1.c++ > ...
#include <iostream>
using namespace std;

class Node {
public:
    int data;
    Node* next;
};

void push(Node** head_ref, int new_data) {
    Node* new_node = new Node();
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}

int getCount(Node* head) {
    int count = 0;
    Node* current = head;
    while (current != NULL) {
        count++;
        current = current->next;
    }
    return count;
}

int main() {
    Node* head = NULL;

    push(&head, 1);
    push(&head, 3);
    push(&head, 1);
    push(&head, 2);
    push(&head, 1);

    cout << "count of nodes is " << getCount(head);
    return 0;
}
```

```
Windows PowerShell
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ensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLau
ncher.exe' '--stdin=Microsoft-MIEngine-In-4sdljqat.1ws' '--stdout=Microsoft-M
IEngine-Out-r2vlzrx0.hrd' '--stderr=Microsoft-MIEngine-Error-x2k5jvoc.0uz' '-
-pid=Microsoft-MIEngine-Pid-jmdsnck2.yzq' '--dbgExe=C:\msys64\ucrt64\bin\gdb.
exe' '--interpreter=mi'
count of nodes is 5
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

```
count of nodes is 5
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Dari percobaan di atas, modifikasilah program sehingga tidak hanya dapat menambah node (**push()**), tapi juga mengurangi node (**deleteNode()**). Kemudian lakukan perhitungan kembali jumlah node tersisa!

Kode pemograman:

```
#include <iostream>
using namespace std;

class Node {
public:
    int data;
    Node* next;
};
void push(Node** head_ref, int new_data) {
    Node* new_node = new Node();
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}
void deleteNode(Node** head_ref, int key) {
    Node* temp = *head_ref;
    Node* prev = NULL;
    if (temp != NULL && temp->data == key) {
        *head_ref = temp->next;
```

```
        delete temp;
        return;
    }
    while (temp != NULL && temp->data != key) {
        prev = temp;
        temp = temp->next;
    }
    if (temp == NULL) return;
    prev->next = temp->next;
    delete temp;
}
int getCount(Node* head) {
    int count = 0;
    Node* current = head;
    while (current != NULL) {
        count++;
        current = current->next;
    }
    return count;
}
int main() {
    Node* head = NULL;
    push(&head, 1);
    push(&head, 3);
    push(&head, 1);
    push(&head, 2);
    push(&head, 1);
    cout << "Before deletion, count of nodes is " <<
getCount(head) << endl;
    deleteNode(&head, 3);
    cout << "After deletion, count of nodes is " <<
getCount(head);
    return 0;
}
```

Hasil

Praktikum > P4 6.2.c++ > getCount(Node *)

```cpp
};
void push(Node** head_ref, int new_data) {
    Node* new_node = new Node();
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}
void deleteNode(Node** head_ref, int key) {
    Node* temp = *head_ref;
    Node* prev = NULL;
    if (temp != NULL && temp->data == key) {
        *head_ref = temp->next;
        delete temp;
        return;
    }
    while (temp != NULL && temp->data != key) {
        prev = temp;
        temp = temp->next;
    }
    if (temp == NULL) return;
    prev->next = temp->next;
    delete temp;
}
int getCount(Node* head) {
    int count = 0;
    Node* current = head;
    while (current != NULL) {
        count++;
        current = current->next;
    }
    return count;
}
int main() {
    Node* head = NULL;
    push(&head, 1);
    push(&head, 3);
    push(&head, 1);
    push(&head, 2);
    push(&head, 1);
    cout << "Before deletion, count of nodes is " << getCount(head) << endl;
    deleteNode(&head, 3);
    cout << "After deletion, count of nodes is " << getCount(head);
    return 0;
}
```

```
indows PowerShell
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S D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\ext
ensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLau
ncher.exe' '--stdin=Microsoft-MIEngine-In-vxliruhn.jma' '--stdout=Microsoft-M
IEngine-Out-0rqiqrgb.t2z' '--stderr=Microsoft-MIEngine-Error-aua0qwap.4xq' '-
-pid=Microsoft-MIEngine-Pid-vggm3rba.asj' '--dbgExe=C:\msys64\ucrt64\bin\gdb.
exe' '--interpreter=mi'
Before deletion, count of nodes is 5
After deletion, count of nodes is 4
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Ln 39, Col 2 Spaces: 4 UTF-8 CRLF {} C++ Go Live Win32

```
Before deletion, count of nodes is 5
After deletion, count of nodes is 4
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

b. Recursive

```
// Recursive C program to find length or count of nodes in a linked list
#include<stdio.h>
#include<stdlib.h>

/* Link list node */
struct Node
{
	int data;
	struct Node* next;
};

/* Given a reference (pointer to pointer) to the head
of a list and an int, push a new node on the front
of the list. */
void push(struct Node** head_ref, int new_data)
{
	/* allocate node */
	struct Node* new_node =
				(struct Node*) malloc(sizeof(struct Node));

	/* put in the data */
	new_node->data = new_data;

	/* link the old list off the new node */
	new_node->next = (*head_ref);

	/* move the head to point to the new node */
```

```
	(*head_ref) = new_node;
}

/* Counts the no. of occurrences of a node
(search_for) in a linked list (head)*/
int getCount(struct Node* head)
{
	// Base case
	if (head == NULL)
		return 0;

	// count is 1 + count of remaining list
	return 1 + getCount(head->next);
}

/* Driver program to test count function*/
int main()
{
	/* Start with the empty list */
	struct Node* head = NULL;

	/* Use push() to construct below list
	1->2->1->3->1 */
	push(&head, 1);
	push(&head, 3);
	push(&head, 1);
	push(&head, 2);
	push(&head, 1);

	/* Check the count function */
	printf("count of nodes is %d", getCount(head));
	return 0;
}
```

Hasil:

```
Praktikum > P4 6.3.cpp > ...
#include<stdio.h>
#include<stdlib.h>

struct Node {
    int data;
    struct Node* next;
};

void push(struct Node** head_ref, int new_data) {
    struct Node* new_node = (struct Node*) malloc(sizeof(struct Node));
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}

int getCount(struct Node* head) {
    if (head == NULL)
        return 0;
    return 1 + getCount(head->next);
}

int main() {
    struct Node* head = NULL;

    push(&head, 1);
    push(&head, 3);
    push(&head, 1);
    push(&head, 2);
    push(&head, 1);

    printf("Count of nodes is %d\n", getCount(head));
    return 0;
}
```

```
Windows PowerShell
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PS D:\1. KULIAH PENS\Semester II\Progdas 2> & 'c:\Users\MyLaptop\.vscode\ext
ensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLau
ncher.exe' '--stdin=Microsoft-MIEngine-In-zx4sgil.4dp' '--stdout=Microsoft-M
IEngine-Out-3gjnyipx.rab' '--stderr=Microsoft-MIEngine-Error-q5itfhe4.0pj' '
-pid=Microsoft-MIEngine-Pid-gvpdcvyj.irq' '--dbgExe=C:\msys64\ucrt64\bin\gdb.
exe' '--interpreter=mi'
Count of nodes is 5
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

```
Count of nodes is 5
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Dari percobaan di atas, modifikasilah program sehingga tidak hanya dapat menambah node (**push()**), tapi juga mengurangi node (**deleteNode()**). Kemudian lakukan perhitungan kembali jumlah node tersisa!

Kode pemograman:

```
#include<stdio.h>
#include<stdlib.h>

struct Node {
    int data;
    struct Node* next;
};
void push(struct Node** head_ref, int new_data) {
    struct Node* new_node = (struct Node*) malloc(sizeof(struct
Node));
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}
void deleteNode(struct Node** head_ref, int key) {
    struct Node* temp = *head_ref;
    struct Node* prev = NULL;
    if (temp != NULL && temp->data == key) {
        *head_ref = temp->next;
        free(temp);
        return;
    }
    while (temp != NULL && temp->data != key) {
        prev = temp;
```

```
        temp = temp->next;
    }
    if (temp == NULL)
        return;
    prev->next = temp->next;
    free(temp);
}

int getCount(struct Node* head) {
    int count = 0;
    struct Node* current = head;
    while (current != NULL) {
        count++;
        current = current->next;
    }
    return count;
}
int main() {
    struct Node* head = NULL;
    push(&head, 1);
    push(&head, 3);
    push(&head, 1);
    push(&head, 2);
    push(&head, 1);
    printf("Count of nodes before deletion : %d\n",
getCount(head));
    int key = 3;
    deleteNode(&head, key);
    printf("Count of nodes after deletion of %d: %d\n", key,
getCount(head));

    return 0;
}
```

Hasil:

Praktikum > P4 6.4.cpp > deleteNode(Node **, int)

```c
void pusn(struct Node** head_ref, int new_data) {
}
void deleteNode(struct Node** head_ref, int key) {
    struct Node* temp = *head_ref;
    Click to add a breakpoint. de* prev = NULL;
    if (temp != NULL && temp->data == key) {
        *head_ref = temp->next;
        free(temp);
        return;
    }
    while (temp != NULL && temp->data != key) {
        prev = temp;
        temp = temp->next;
    }
    if (temp == NULL)
        return;
    prev->next = temp->next;
    free(temp);
}

int getCount(struct Node* head) {
    int count = 0;
    struct Node* current = head;
    while (current != NULL) {
        count++;
        current = current->next;
    }
    return count;
}
int main() {
    struct Node* head = NULL;
    push(&head, 1);
    push(&head, 3);
    push(&head, 1);
    push(&head, 2);
    push(&head, 1);
    printf("Count of nodes before deletion : %d\n", getCount(head));
    int key = 3;
    deleteNode(&head, key);
    printf("Count of nodes after deletion of %d: %d\n", key, getCount(head));

    return 0;
}
```

```
Windows PowerShell
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PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-vkekzco3.yot' '--stdout=Microsoft-MIEngine-Out-ucrzqlos.cqw' '--stderr=Microsoft-MIEngine-Error-msfi22in.mlc' '--pid=Microsoft-MIEngine-Pid-2gjlpohe.nar' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Count of nodes before deletion : 5
Count of nodes after deletion of 3: 4
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

```
Count of nodes before deletion : 5
Count of nodes after deletion of 3: 4
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

7. **Percobaan 7 – Mencari Sebuah Elemen di Dalam Singly Linked List**

   Pada percobaan ini, elemen dari sebuah SLL dapat dicari dengan menggunakan key/nilai datanya. Program akan menunjukkan “Ya” jika data tersebut ada dalam SLL dan “No” jika data tidak tersedia di dalam SLL. Pencarian dapat dilakukan dengan menggunakan 2 teknik, yaitu iterative dan recursive.

a. Iterative

```
// Iterative C++ program to search
// an element in linked list
#include <bits/stdc++.h>
using namespace std;

/* Link list node */
class Node
{
	public:
	int key;
	Node* next;
};

/* Given a reference (pointer to pointer) to the head
of a list and an int, push a new node on the front
of the list. */
void push(Node** head_ref, int new_key)
{
	/* allocate node */
	Node* new_node = new Node();

	/* put in the key */
	new_node->key = new_key;

	/* link the old list off the new node */
	new_node->next = (*head_ref);

	/* move the head to point to the new node */
	(*head_ref) = new_node;
}

/* Checks whether the value x is present in linked list */
bool search(Node* head, int x)
{
	Node* current = head; // Initialize current
	while (current != NULL)
	{
```

```
                if (current->key == x)
                        return true;
                current = current->next;
        }
        return false;
}

/* Driver program to test count function*/
int main()
{
        /* Start with the empty list */
        Node* head = NULL;
        int x = 21;

        /* Use push() to construct below list
        14->21->11->30->10 */
        push(&head, 10);
        push(&head, 30);
        push(&head, 11);
        push(&head, 21);
        push(&head, 14);

        search(head, 21)? cout<<"Yes" : cout<<"No";
        return 0;
}
```

Hasil percobaan:

Praktikum > P4 7.1.cpp > ...

```
#include <iostream>
using namespace std;

class Node {
public:
    int key;
    Node* next;
};

void push(Node** head_ref, int new_key) {
    Node* new_node = new Node();
    new_node->key = new_key;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}

bool search(Node* head, int x) {
    Node* current = head;
    while (current != NULL) {
        if (current->key == x)
            return true;
        current = current->next;
    }
    return false;
}

int main() {
    Node* head = NULL;
    int x = 21;

    push(&head, 10);
    push(&head, 30);
    push(&head, 11);
    push(&head, 21);
    push(&head, 14);

    search(head, 21)? cout << "Yes" : cout << "No";
    return 0;
}
```

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-au4lmjv3.24n' '--stdout=Microsoft-MIEngine-Out-3s0bnnyw.cyj' '--stderr=Microsoft-MIEngine-Error-i0ct0rrb.esm' '--pid=Microsoft-MIEngine-Pid-jqnucsdk.cdt' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Yes
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

```
Yes
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Dari percobaan di atas, cobalah mencari data lainnya! Data yang anda masukkan boleh acak (bisa ada dan bisa tidak dari data SLL).

Mencari data 14

```
    search(head, 14)? cout << "Yes" : cout << "No";
```

```
Yes
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Mencari data 92

```
    search(head, 92)? cout << "Yes" : cout << "No";
```

```
No
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

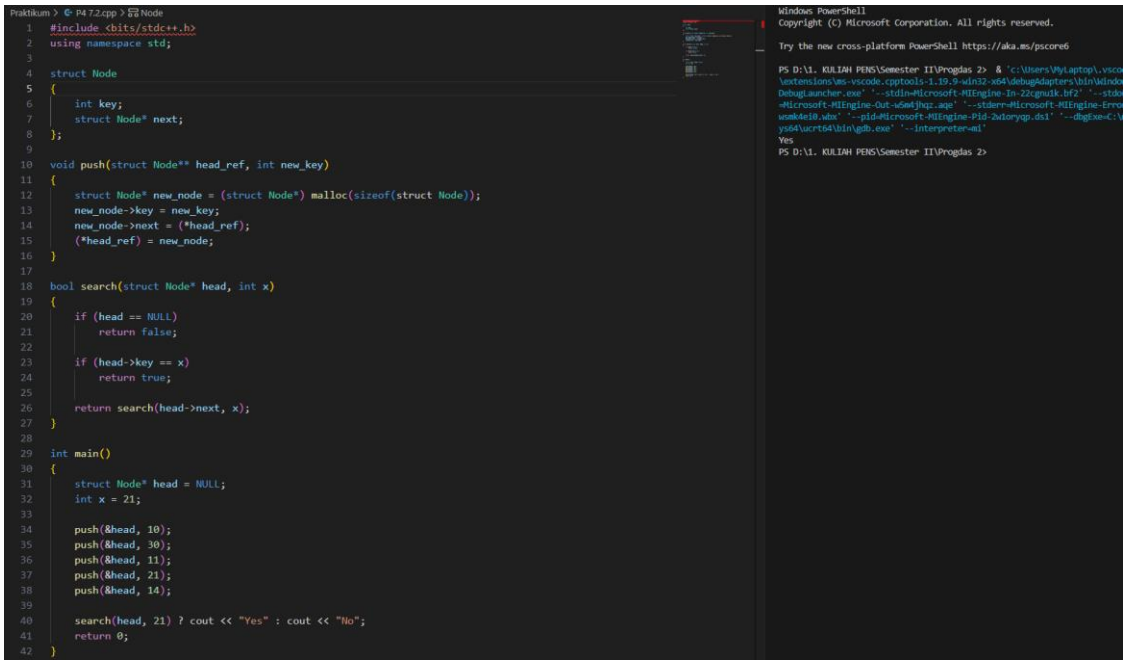
b. Recursive

```
// Recursive C++ program to search
// an element in linked list
#include <bits/stdc++.h>
using namespace std;

/* Link list node */
struct Node
{
        int key;
        struct Node* next;
};

/* Given a reference (pointer to pointer) to the head
```

```
of a list and an int, push a new node on the front
of the list. */
void push(struct Node** head_ref, int new_key)
{
	/* allocate node */
	struct Node* new_node = (struct Node*) malloc(sizeof(struct Node));

	/* put in the key */
	new_node->key = new_key;

	/* link the old list off the new node */
	new_node->next = (*head_ref);

	/* move the head to point to the new node */
	(*head_ref) = new_node;
}

/* Checks whether the value x is present in linked list */
bool search(struct Node* head, int x)
{
	// Base case
	if (head == NULL)
		return false;

	// If key is present in current node, return true
	if (head->key == x)
		return true;

	// Recur for remaining list
	return search(head->next, x);
}

/* Driver code*/
int main()
{
	/* Start with the empty list */
	struct Node* head = NULL;
	int x = 21;

	/* Use push() to construct below list
	14->21->11->30->10 */
	push(&head, 10);
	push(&head, 30);
	push(&head, 11);
	push(&head, 21);
	push(&head, 14);
```

```
        search(head, 21)? cout << "Yes" : cout << "No";
        return 0;
}
```

Hasil percobaan:




```
Yes
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Dari percobaan di atas, cobalah mencari data lainnya! Data yang anda masukkan bolehacak (bisa ada dan bisa tidak dari data SLL).

Mencari data 39

```
    search(head, 39) ? cout << "Yes" : cout << "No";
```

```
Yes
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Mencari data 21

```
    search(head, 21) ? cout << "Yes" : cout << "No";
```

```
No
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

c. **Tugas:** Dari kedua Teknik pencarian key/nilai data di atas, manakah yang lebih cepat? Menggunakan Teknik iterative atau menggunakan Teknik recursive? Tunjukkan dalam waktu (millisecond)!

Anda boleh menggunakan library chrono seperti dibawah ini:

```
#include <iostream>
#include <chrono>

using namespace std;
int main()
{
   chrono::steady_clock sc;   // create an object of `steady_clock` class
   auto start = sc.now();     // start timer

   // do stuff....

   auto end = sc.now();       // end timer (starting & ending is done by measuring the time
   auto time_span = static_cast<chrono::duration<double>>(end - start);   // measure time s
   cout<<"Operation took: "<<time_span.count()<<" seconds !!!";
   return 0;
}
```

Atau menggunakan library ctime seperti di bawah ini:

```
#include <iostream>
#include <cstdio>
#include <ctime>

int main()
{
    std::clock_t start;
    double duration;

    start = std::clock(); // get current time

    // Do your stuff here

    duration = ( std::clock() - start ) / (double) CLOCKS_PER_SEC;

    std::cout << "Operation took "<< duration << "seconds" << std::endl;

    return 0;
}
```

Di dalam “do stuff” adalah fungsi search(head,X). Artinya, variable start–duration atau start–end mengapit fungsi search(head, X) saja.

Kode pemograman

```
#include <iostream>
#include <chrono>
#include <cstdlib>
using namespace std;

struct Node {
    int data;
    struct Node* next;
};
```

```
void push(struct Node** head_ref, int new_data) {
    struct Node* new_node = new Node();
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}

bool searchIterative(struct Node* head, int x) {
    struct Node* current = head;
    while (current != NULL) {
        if (current->data == x)
            return true;
        current = current->next;
    }
    return false;
}

bool searchRecursive(struct Node* head, int x) {
    if (head == NULL)
        return false;
    if (head->data == x)
        return true;
    return searchRecursive(head->next, x);
}

int main() {
    struct Node* head = NULL;
    srand(time(NULL));
    for (int i = 0; i < 10000; i++) {
        push(&head, rand() % 1000);
    }
    int target = rand() % 1000;

    auto startIterative = chrono::high_resolution_clock::now();
    searchIterative(head, target);
    auto endIterative = chrono::high_resolution_clock::now();
    chrono::duration<double, nano> durationIterative =
endIterative - startIterative;
    auto startRecursive = chrono::high_resolution_clock::now();
    searchRecursive(head, target);
    auto endRecursive = chrono::high_resolution_clock::now();
    chrono::duration<double, nano> durationRecursive =
endRecursive - startRecursive;
    cout << "Nilai yang dicari ialah: " << target << endl;
```

```
    cout << "Waktu pencarian menggunakan iterative: " <<
durationIterative.count() << " ns" << endl;
    cout << "Waktu pencarian menggunakan recursive: " <<
durationRecursive.count() << " ns" << endl;
    return 0;
}
```

Hasil dalam 10x percobaan yang berbeda:

```
DebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-zfzc2p3j.vyq' '--stdout
ektotojz.txn' '--pid=Microsoft-MIEngine-Pid-delxsgm2.aml' '--dbgExe=C:\ms
ys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 820
Waktu pencarian menggunakan iterative: 600 ns
Waktu pencarian menggunakan recursive: 400 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode
\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\Windows
DebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-yrlbnoxj.whi' '--stdout
hi3iktsq.n1z' '--pid=Microsoft-MIEngine-Pid-oyco1wtu.1x3' '--dbgExe=C:\ms
ys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 436
Waktu pencarian menggunakan iterative: 12600 ns
Waktu pencarian menggunakan recursive: 120000 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-
9.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-v5pqw5g5.rvz' '
tdout=Microsoft-MIEngine-Out-xpiv55yt.zw3' '--stderr=Microsoft-MIEngine-Error-bwwqnoni.3zw' '--pid=Micro
t-MIEngine-Pid-fcq4kqd3.y1x' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 480
Waktu pencarian menggunakan iterative: 7800 ns
Waktu pencarian menggunakan recursive: 80600 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-
9.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-2kzib4fd.0o3' '
tdout=Microsoft-MIEngine-Out-u2jopf5u.sk0' '--stderr=Microsoft-MIEngine-Error-125fdqqh.5na' '--pid=Micro
t-MIEngine-Pid-dnahx3cm.h3y' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 533
Waktu pencarian menggunakan iterative: 5400 ns
Waktu pencarian menggunakan recursive: 33800 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-
9.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-j4t1bqtq.hqm' '
tdout=Microsoft-MIEngine-Out-2hfbqgfa.11j' '--stderr=Microsoft-MIEngine-Error-euhxp2wi.ppx' '--pid=Micro
t-MIEngine-Pid-405pveqt.ftl' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 744
Waktu pencarian menggunakan iterative: 5600 ns
Waktu pencarian menggunakan recursive: 51000 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-
9.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-k0mrhz40.zxl' '
tdout=Microsoft-MIEngine-Out-bhbgb4na.ar3' '--stderr=Microsoft-MIEngine-Error-wlfa5pba.5lm' '--pid=Micro
t-MIEngine-Pid-jqatkprm.qhr' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 788
Waktu pencarian menggunakan iterative: 14200 ns
Waktu pencarian menggunakan recursive: 119500 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-
9.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-ilwh1tcn.uer' '
tdout=Microsoft-MIEngine-Out-lwfcdyet.3qc' '--stderr=Microsoft-MIEngine-Error-4gp0xknw.2pw' '--pid=Micro
t-MIEngine-Pid-wr02h3dg.vpd' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 599
Waktu pencarian menggunakan iterative: 8800 ns
Waktu pencarian menggunakan recursive: 89500 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>  & 'c:\Users\MyLaptop\.vscode\extensions\ms-vscode.cpptools-
9.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-yufy5fvd.oyd' '
tdout=Microsoft-MIEngine-Out-lvhgqgi0.x02' '--stderr=Microsoft-MIEngine-Error-l2aebu1k.vph' '--pid=Micro
t-MIEngine-Pid-5h0dvaea.r0d' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Nilai yang dicari ialah: 325
Waktu pencarian menggunakan iterative: 4400 ns
Waktu pencarian menggunakan recursive: 34100 ns
PS D:\1. KULIAH PENS\Semester II\Progdas 2>
```

Dari percobaan ini adalah bahwa secara umum, metode pencarian iterative lebih cepat daripada metode pencarian recursive untuk linked list. Meskipun ada beberapa kasus di mana metode recursive memiliki waktu yang sedikit lebih cepat atau sama dengan iterative, namun perbedaan ini jarang terjadi dan iterative cenderung lebih efisien dalam hal waktu pencarian. Faktor-faktor seperti struktur linked list, ukuran data, dan nilai-nilai yang dicari juga dapat mempengaruhi hasil pengukuran waktu.

8. Percobaan 8 – Mendapatkan Node ke-N dari Singly Linked List

Pada percobaan ini, nilai data dari node sebuah SLL dapat diketahui dengan menggunakan fungsi GetNth(head, 3). Program akan menunjukkan nilai data pada nomor node yang diberikan. Pencarian dapat dilakukan dengan menggunakan 2 teknik, yaitu iterative dan recursive.

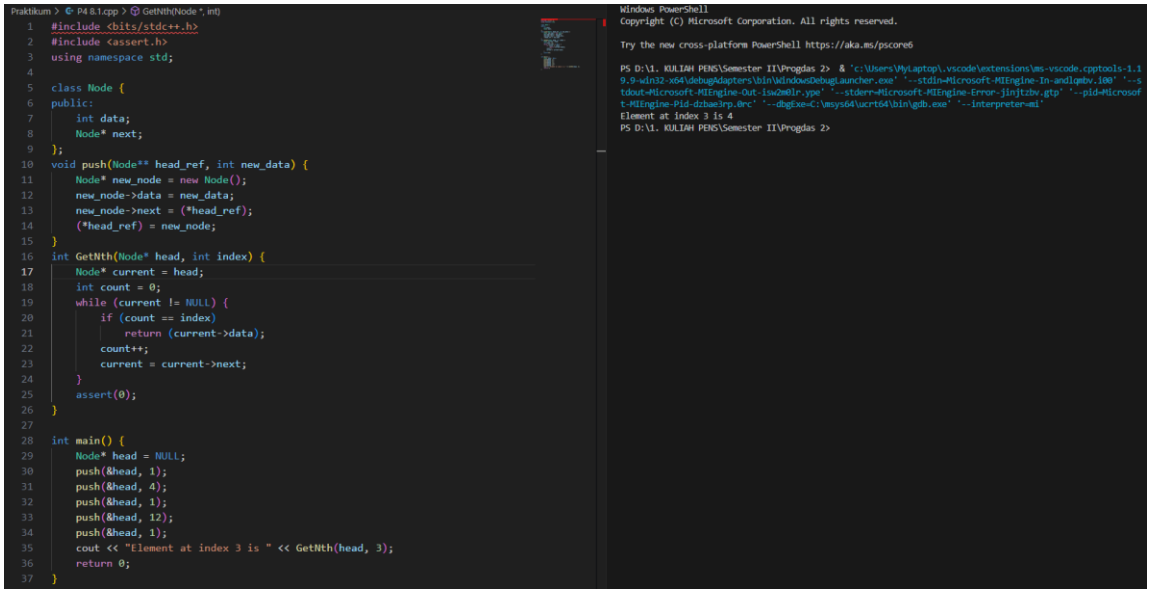
a. Menggunakan Fungsi Iterative

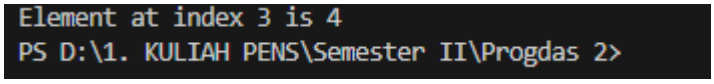
```
// C++ program to find n'th
// node in linked list
#include <bits/stdc++.h>
#include <assert.h>
using namespace std;

// Link list node
class Node
{
	public:
	int data;
	Node* next;
};

/* Given a reference (pointer to
pointer) to the head of a list
and an int, push a new node on
the front of the list. */
void push(Node** head_ref, int new_data)
{

	// allocate node
	Node* new_node = new Node();

	// put in the data
	new_node->data = new_data;

	// link the old list
	// off the new node
	new_node->next = (*head_ref);

	// move the head to point
	// to the new node
	(*head_ref) = new_node;
```

```
}

// Takes head pointer of
// the linked list and index
// as arguments and return
// data at index
int GetNth(Node* head, int index)
{

	Node* current = head;

	// the index of the
	// node we're currently
	// looking at
	int count = 0;
	while (current != NULL)
	{
		if (count == index)
			return(current->data);
		count++;
		current = current->next;
	}

	/* if we get to this line,
	the caller was asking
	for a non-existent element
	so we assert fail */
	assert(0);
}

// Driver Code
int main()
{

	// Start with the
	// empty list
	Node* head = NULL;

	// Use push() to construct
	// below list
	// 1->12->1->4->1
```

```
        push(&head, 1);
        push(&head, 4);
        push(&head, 1);
        push(&head, 12);
        push(&head, 1);

        // Check the count
        // function
        cout << "Element at index 3 is " << GetNth(head, 3);
        return 0;
}
```

Hasil percobaan:





Dari percobaan di atas, cobalah mencari data lainnya! Nomor node yang anda masukkan boleh acak (bisa ada dan bisa tidak dari data SLL).

b. Menggunakan Fungsi Recursive

```
// C program to find n'th node in linked list
// using recursion
#include <bits/stdc++.h>
using namespace std;

/* Link list node */
struct Node
{
int data;
struct Node* next;
};

/* Given a reference (pointer to pointer) to
the head of a list and an int, push a
new node on the front of the list. */
void push(struct Node** head_ref, int new_data)
{
/* allocate node */
struct Node* new_node =
                    (struct Node*) malloc(sizeof(struct Node));

/* put in the data */
new_node->data = new_data;

/* link the old list off the new node */
new_node->next = (*head_ref);

/* move the head to point to the new node */
(*head_ref) = new_node;
```

```
}

/* Takes head pointer of the linked list and index
 as arguments and return data at index*/
int GetNth(struct Node *head,int n)
{
 int count = 1;

 //if count equal too n return node->data
 if(count == n)
 return head->data;

 //recursively decrease n and increase
 // head to next pointer
 return GetNth(head->next, n-1);
}

/* Drier program to test above function*/
int main()
{
 /* Start with the empty list */
 struct Node* head = NULL;

 /* Use push() to construct below list
 1->12->1->4->1 */
 push(&head, 1);
 push(&head, 4);
 push(&head, 1);
 push(&head, 12);
 push(&head, 1);

 /* Check the count function */
 printf("Element at index 3 is %d", GetNth(head, 3));
 getchar();
}
```

Hasil percobaan:

```
Praktikum > P4 8.2.cpp > ...
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* next;
};

void push(struct Node** head_ref, int new_data) {
    struct Node* new_node = (struct Node*) malloc(sizeof(struct Node));
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}

int GetNth(struct Node* head, int n) {
    int count = 1;
    if (count == n)
        return head->data;
    return GetNth(head->next, n - 1);
}

int main() {
    struct Node* head = NULL;

    push(&head, 1);
    push(&head, 4);
    push(&head, 1);
    push(&head, 12);
    push(&head, 1);

    printf("Element at index 3 is %d", GetNth(head, 3));
    getchar();
}
```

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS D:\1. KULIAH PENS\Semester II\Progdas 2> & 'c:\Users\MyLaptop\
.vscode\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapte
rs\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-uq
rkqrhk.dbq' '--stdout=Microsoft-MIEngine-Out-xvprrkay.vmx' '--stde
rr=Microsoft-MIEngine-Error-eebet0kp.r0h' '--pid=Microsoft-MIEngin
e-Pid-nzscgn3d.y2e' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--int
erpreter=mi'
Element at index 3 is 1
```

```
Element at index 3 is 1
```

Dari percobaan di atas, cobalah mencari data lainnya! Nomor node yang anda masukkan boleh acak (bisa ada dan bisa tidak dari data SLL).

Nomor node 4

```
    printf("Element at index 4 is %d", GetNth(head, 4));
```

```
Element at index 4 is 4
```

Nomor node 5

```
    printf("Element at index 5 is %d", GetNth(head, 5));
```

```
Element at index 5 is 1
```

Nomor node 10

```
    printf("Element at index 10 is %d", GetNth(head, 10));
```

```
WATCH

16  int GetNth(struct Node* head, int n) {
17      int count = 1;
18      if (count == n)
19          return head->data;
20      return GetNth(head->next, n - 1);

Exception has occurred. ×
Segmentation fault

21  }
22
23  int main() {
24      struct Node* head = NULL;
25
26      push(&head, 1);
27      push(&head, 4);
28      push(&head, 1);
29      push(&head, 12);
30      push(&head, 1);
31
32      printf("Element at index 10 is %d", GetNth(head, 10));
33      getchar();
34  }

CALL STACK
PAUSED ON EXCEPTION
GetNth(Node * head
GetNth(Node * head
GetNth(Node * head
GetNth(Node * head
GetNth(Node * head
GetNth(Node * head
main()  P4 8.2.c...
```

Dikarenakan nomor node tersebut tidaklah ada